Benchmark: **MA.7.GR.2.3** · Volume of a cylinder Student code: _____ Calculator: yes

Learning goal: I can find the volume of a right circular cylinder. (circle your level after scoring)

4	I can solve a real-world volume problem, including comparing two cylinders.
3	I can find the volume of a cylinder from its radius or diameter and height.
2	I can find the area of the base.
1	I need help to start this benchmark.

1. A cylindrical tank has a radius of 8 feet and a height of 12 feet.

What is the volume of the tank? Round your answer to the nearest hundredth. Use 3.14 for π.

2. A cylindrical container has a diameter of 16 inches and a height of 24 inches.

What is the volume of the container? Write your answer in terms of π.

Turn over →

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder Student code: _____

3. A right circular cylinder has a radius of 6 units and a height of 16 units.

What is the volume of the cylinder, in cubic units?

A. 192π

B. 1152π

C. 576π

D. 96π

4. A cylindrical trash can has a diameter of 20 inches and a height of 26 inches.

What is the volume of the trash can? Round your answer to the nearest hundredth. Use 3.14 for π.

5. A cylindrical storage tank has a volume of 478 cubic feet and a radius of 3.2 feet.

What is the height of the storage tank? Round your answer to the nearest hundredth. Use the π key on your calculator.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.2.3 · MA.7.GR.2.3 · Volume of a cylinder

Unit 12: Surface Area and Volume. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **2411.52 cubic feet**

$V = \pi r^2 h = \pi(8)^2(12) = 768\pi$.

With 3.14 for π: $768 \times 3.14 = 2411.52$ cubic feet.

2. Regular **1536π cubic inches**

The radius is $16 \div 2 = 8$ inches.

$V = \pi r^2 h = \pi(8)^2(24) = 1536\pi$ cubic inches.

3. Regular **C. 576π**

$V = \pi r^2 h = \pi(6)^2(16) = 576\pi$.

The value 192π would be the curved surface area instead.

4. Regular **8164 cubic inches (8164.00 also accepted)**

The radius is $20 \div 2 = 10$ inches.

$V = \pi r^2 h = \pi(10)^2(26) = 2600\pi$.

With 3.14 for π: $2600 \times 3.14 = 8164.00$ cubic inches.

5. Rigor **14.86 feet**

$V = \pi r^2 h$, so $478 = \pi(3.2)^2 h = 10.24\pi h$.

$h = 478 \div (10.24\pi) \approx 14.86$ feet with the π key.

Re-check strip (3 items, C-GR.2.3-R)

1. Regular **C. 256π**

$V = \pi r^2 h = \pi(8)^2(4) = 256\pi$.

The value 64π would be the curved surface area instead.

2. Regular **339.29 cubic feet**

The radius is $12 \div 2 = 6$ feet.

$V = \pi r^2 h = \pi(6)^2(3) = 108\pi$.

With the π key: $108 \times \pi \approx 339.29$ cubic feet.

3. Regular **384π cubic inches**

The radius is $8 \div 2 = 4$ inches.

$V = \pi r^2 h = \pi(4)^2(24) = 384\pi$ cubic inches.

Comparison strip (2 items, C-GR.2.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **1005.31 cubic feet**

$V = \pi r^2 h = \pi(8)^2(5) = 320\pi$.

With the π key: $320 \times \pi \approx 1005.31$ cubic feet.

2. Regular **64π cubic inches**

The radius is $4 \div 2 = 2$ inches.

$V = \pi r^2 h = \pi(2)^2(16) = 64\pi$ cubic inches.

Reteach plan — C-GR.2.3

MA.7.GR.2.3 · Volume of a cylinder

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items)

C-GR.2.3-R

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder

Student code: _____ Calculator: yes

1. A right circular cylinder has a radius of 8 units and a height of 4 units.

What is the volume of the cylinder, in cubic units?

A. 64π

B. 32π

C. 256π

D. 512π

2. A cylindrical swimming pool has a diameter of 12 feet and a height of 3 feet.

What is the volume of the pool? Round your answer to the nearest hundredth. Use the π key on your calculator.

3. A cylindrical container has a diameter of 8 inches and a height of 24 inches.

What is the volume of the container? Write your answer in terms of π.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items)

C-GR.2.3-R

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder

Student code: _____ Calculator: yes

1. A right circular cylinder has a radius of 8 units and a height of 4 units.

What is the volume of the cylinder, in cubic units?

A. 64π

B. 32π

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2. A cylindrical swimming pool has a diameter of 12 feet and a height of 3 feet.

What is the volume of the pool? Round your answer to the nearest hundredth. Use the π key on your calculator.

3. A cylindrical container has a diameter of 8 inches and a height of 24 inches.

What is the volume of the container? Write your answer in terms of π.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-GR.2.3-X

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder

Student code: _____ Calculator: yes

1. A cylindrical tank has a radius of 8 feet and a height of 5 feet.

What is the volume of the tank? Round your answer to the nearest hundredth. Use the π key on your calculator.

2. A cylindrical container has a diameter of 4 inches and a height of 16 inches.

What is the volume of the container? Write your answer in terms of π.

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-GR.2.3-X

Benchmark: **MA.7.GR.2.3** · Volume of a cylinder

Student code: _____ Calculator: yes

1. A cylindrical tank has a radius of 8 feet and a height of 5 feet.

What is the volume of the tank? Round your answer to the nearest hundredth. Use the π key on your calculator.

2. A cylindrical container has a diameter of 4 inches and a height of 16 inches.

What is the volume of the container? Write your answer in terms of π.

Score: ____ / 2 → **copy to your progress strip**